

NEAR-EARTH ASTEROID

433 Eros

AVERAGE DISTANCE TO SUN 136 million miles (218 million km)

ORBITAL PERIOD 1.76 years

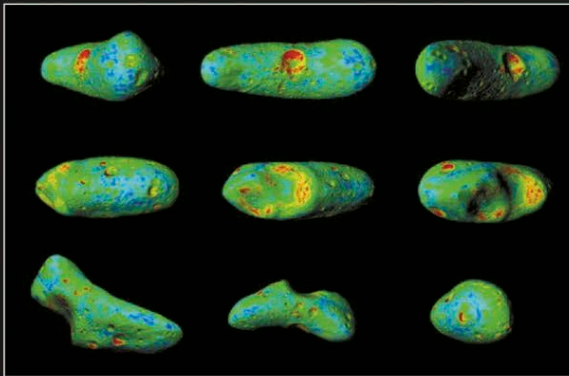
ROTATION PERIOD 5.27 hours

LENGTH 19.25 miles (31 km)

DATE OF DISCOVERY August 13, 1898

Lying in near-Earth orbit, outside the Main Belt, Eros is usually closer to the Sun than to Mars (see p.170). Its orbit also brings it close to Earth; at the last close approach, in 1975, Eros came within 14 million miles (22 million km) of the planet. The orbit is unstable, and Eros has a 1-in-10 chance of hitting either Earth or Mars in the next million years. In 1960, Eros was detected by radar, and infrared measurements taken in the 1970s indicated that the surface was not just bare rock but was covered by a thermally insulating blanket of dust and rock fragments. Eros was the first asteroid to be orbited by a spacecraft and the first to be landed on. It was chosen for close study because it is big and nearby.

On February 14, 2000, the Near Earth Asteroid Rendezvous (NEAR) spacecraft (renamed NEAR Shoemaker in March 2000) went into orbit around Eros. It landed 363 days later. About 160,000 images were recorded. They revealed an irregularly shaped body that had heavily cratered 2-billion-year-old areas lying next to relatively smooth regions. Even though the gravitational



COMPUTER MODELS

The gravity on Eros is about 1/2,000th of that on Earth but varies by nearly a factor of two from place to place. The colors in this image represent the rate at which a rock would roll downhill. It would roll fastest in the red areas, and it wouldn't move at all in the blue areas.

field is very small, several thousand boulders larger than 50 ft (15 m) across have fallen back to the surface after being ejected by impacts, and some surface dust has rolled down the slopes to form sand dunes a few

yards (meters) deep. Laser measurements of the NEAR-Eros distance as the spacecraft orbited have not only produced an accurate map of the asteroid's shape but also indicated that the interior is nearly uniform, with a density about the same as that of Earth's crust. Eros is not a pile of rubble like Mathilde;

it is a single solid lump of rock. The spacecraft's gamma-ray spectrometer worked for 2 weeks after touchdown. Eros was found to be silicate-rich and highly reflective.

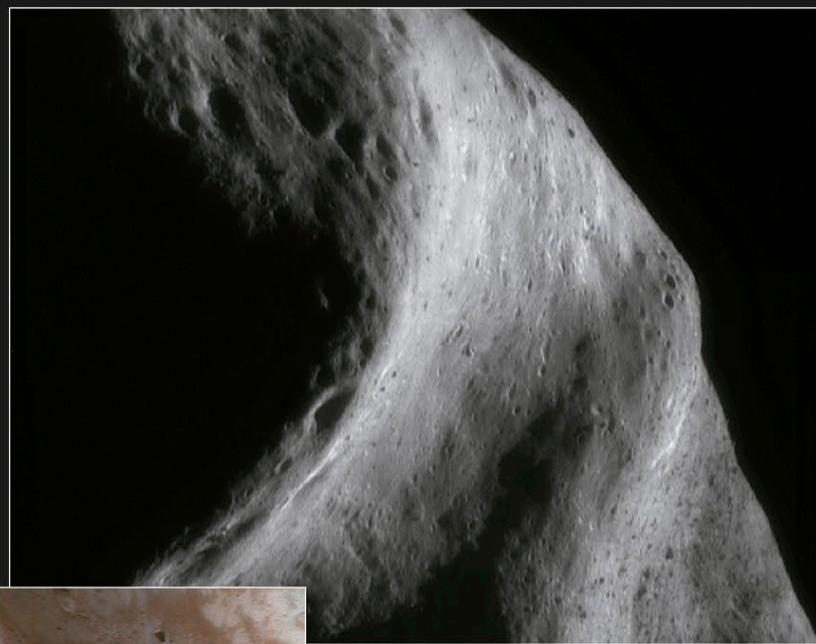
ROCK AND REGOLITH

Some of the rocks and regolith on Eros's surface have a red coloring. The longer their exposure to minor impacts and the solar wind, the redder they appear.



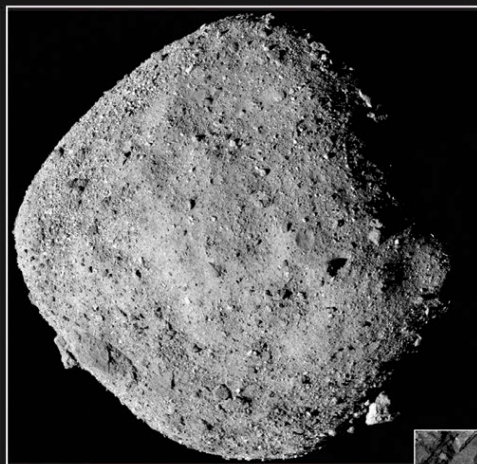
CLOSING IN

NEAR Shoemaker took this image from a height of 3,770 ft (1,150 m) shortly before it touched down on Eros.



THE SADDLE

Four images have been combined to produce this view of the "saddle" region at the south of the asteroid. This 6-mile (10-km) wide scoop, which has been named Himeros, is relatively boulder-free, unlike the region at the lower right of the frame.

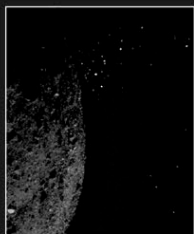


MOSAIC IMAGE OF BENNU

This global view of Bennu was created using 12 images collected from a range of 15 miles (24 km). The "spinning top" shape is clear. The giant boulder at the lower right is about 180 ft (55 m) across.

EJECTING PARTICLES

Several particle ejection events have been seen at Bennu. It is not known whether these are caused by meteoroid impacts, thermal stress fracturing, or release of water vapor.



NEAR-EARTH ASTEROID

101955 Bennu

AVERAGE DISTANCE TO SUN 105 million miles (169 million km)

ORBITAL PERIOD 1.20 years

ROTATION PERIOD 4.30 hours

DIAMETER 0.30 miles (0.49 km)

DATE OF DISCOVERY September 11, 1999

Bennu is the target of NASA's OSIRIS-REx mission, which arrived in orbit around it in December 2018 and is expected to return samples to Earth in 2023. Previously known only by the provisional designation 1999 RQ₃₆, Bennu's name was chosen out of more than 8,000 entries as the winner of a school "name that asteroid" contest after it had been selected as the target for OSIRIS-REx.

CLOSELY PACKED BOULDERS

This view from a range of only 3 miles (5 km) shows jagged boulders of many sizes, some darker than others. The largest here is about 24 ft (7 m) across.

Its shape resembles that of a spinning top, which results from the effect of its rapid rotation on its unconsolidated interior. Despite being rocky, its overall density is only slightly greater than that of water, so it is almost certainly a "rubble pile" with spaces between its internal components. The rubble-pile structure is similar to that of Itokawa (see p.175), but Bennu is rotating about twice as fast and its rotation period is decreasing by about 1 second every 100 years. Its exposure to heat from the Sun while it rotates causes uneven emissions of thermal radiation from its surface, which accelerates its rotation speed. This is a phenomenon known as the YORP (Yarkovsky-O'Keefe-Radzievskii-Paddack) effect.

The way it reflects sunlight suggests that Bennu's composition is similar to carbonaceous chondrite asteroids, but that won't be confirmed until the samples are returned and analyzed.

Bennu's orbit crosses the Earth's, and it is estimated to have a 1-in-2,700 chance of colliding with the Earth between 2175 and 2199.